7 **(a)** Fig. 7.1 is a scale drawing of light waves approaching a narrow slit.

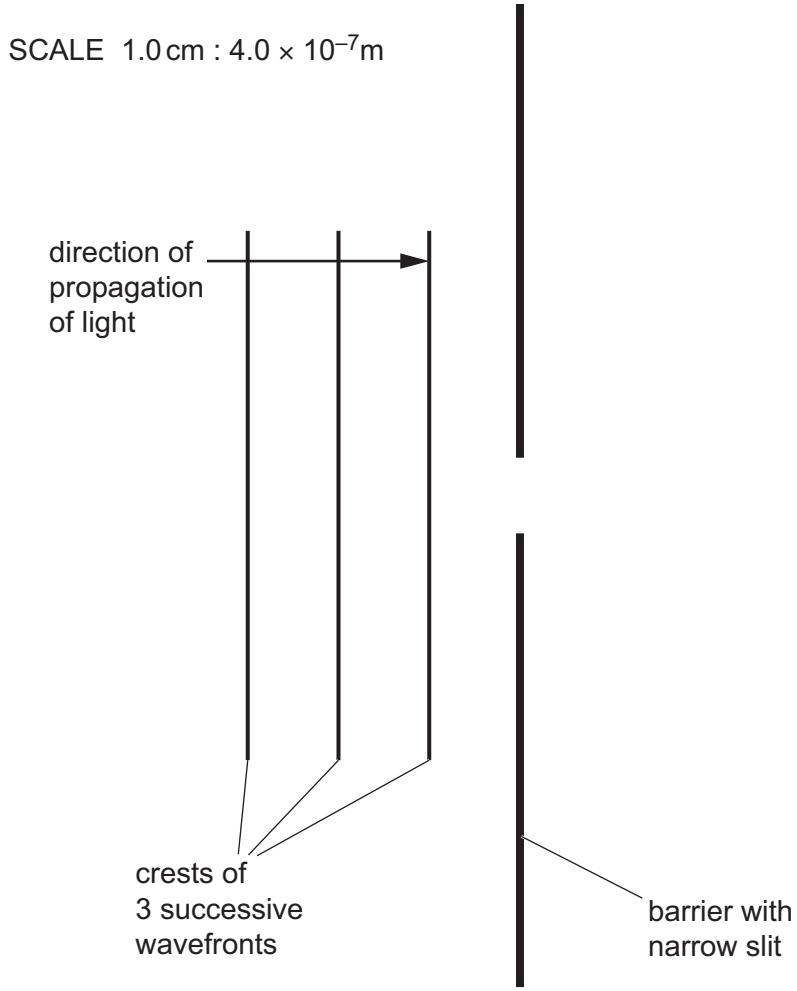

Fig. 7.1

(i) Name the wave effect produced by the narrow slit.

..... [1]

(ii) Using Fig. 7.1, determine the wavelength of the light. Give your answer to **two** significant figures.

wavelength = ... [2]

(iii) On Fig. 7.1, draw **three** wavefronts that have passed through the narrow slit. [3]

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(b) A foghorn emits a sound with frequency 380 Hz. The sound is heard by a ship 2.5 km away from the foghorn. The speed of sound in air is 330 m / s.

(i) Show that the wavelength of the sound is approximately 0.9 m.
State any equation you use in words or symbols.

[2]

(ii) Calculate the time it takes for sound to travel to the ship from the foghorn.

time = ... [2]

[Total: 10]

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